LABORATORY PROJECT: THE FOUR-TANK PROCESS

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1. Abstract

The following report is part of EL2520 final project. It investigates the dynamics of a Four-Tank system. In the report, the parameters describing the system are experimentally calculated. Then, the system is controlled with a dynamic decoupling controller as well as its robustified version (Glover-McFarlane) for both the minimum and non-minimum phase cases. The former controller showed better resistance to higher disturbances, whereas the latter proved better at reference tracking.

2. Laboratory Occasion 1

2.1. **Modeling.**

2.1.1. *Getting the system equations.* We will deduce the equations of the system tank by tank. Tank 1 receives an input from the pump 1 quantified by the equation $q_{inpump} = \gamma_1 k_1 u_1$ since the flux is divided with coefficient γ_1. It also receives an input from the tank above, tank 3, from its hole of dimensions a_3. This can be quantified with Bernoulli's law: $q_{3,out} = a_3\sqrt{2gh_3}$. Finally, tank 1 loses water from its own hole following the equation $q_{1,out} = a_1\sqrt{2gh_1}$. Once this is done, we simply use the relation between the cross section area, the height of the water and the debit going in and out. from this we deduce:

$$
\begin{aligned}
A_1\frac{dh_1}{dt} &= -q_{1,out} + q_{1,in} + q_{3,out} \\
\Leftrightarrow \frac{dh_1}{dt} &= -\frac{a_1}{A_1}\sqrt{2gh_1} + \frac{a_3}{A_1} + \frac{\gamma_1 k_1}{A_1}u_1
\end{aligned}
$$

We do a similar analysis for each of the tanks. Tank 2 receives from tank 4, the pump with a coefficient γ_2 and also loses some water. Tank 3 receive from pump 1 with a coefficient $1-\gamma_1$, and lose water in one, tank 4 receive from pump 2 with coefficient $1-\gamma_2$ and lose some water in tank 2. From this analysis we deduce the equation system:

$$\frac{dh_1}{dt} = -\frac{a_1}{A_1}\sqrt{2gh_1} + \frac{a_3}{A_1}\sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1}u_1 \tag{2.1}$$

$$\frac{dh_2}{dt} = -\frac{a_2}{A_2}\sqrt{2gh_2} + \frac{a_4}{A_2}\sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2}u_2 \tag{2.2}$$

$$\frac{dh_3}{dt} = -\frac{a_3}{A_3}\sqrt{2gh_3} + \frac{(1-\gamma_1)k_1}{A_3}u_1 \tag{2.3}$$

$$\frac{dh_4}{dt} = -\frac{a_4}{A_4}\sqrt{2gh_4} + \frac{(1-\gamma_2)k_2}{A_4}u_2 \tag{2.4}$$

2.1.2. *Equilibrium equation.* At equilibrium, $\frac{dh_i}{dt} = 0$. Hence we get this system of equation:

$$0 = -a_1\sqrt{2gh_1^0} + a_3\sqrt{2gh_3^0} + \gamma_1 k_1 u_1^0$$
$$0 = -a_2\sqrt{2gh_2^0} + a_4\sqrt{2gh_4^0} + \gamma_2 k_2 u_2^0$$
$$0 = -a_3\sqrt{2gh_3^0} + (1-\gamma_1) k_1 u_1^0$$
$$0 = -a_4\sqrt{2gh_4^0} + (1-\gamma_2) k_2 u_2^0$$

We then replace h_i^0 with $\frac{y_i}{k_c}$

$$0 = -\frac{a_1}{\sqrt{k_c}}\sqrt{2gy_1^0} + \frac{a_3}{\sqrt{k_c}}\sqrt{2gy_3^0} + \gamma_1 k_1 u_1^0$$
$$0 = -\frac{a_2}{\sqrt{k_c}}\sqrt{2gy_2^0} + \frac{a_4}{\sqrt{k_c}}\sqrt{2gy_4^0} + \gamma_2 k_2 u_2^0$$
$$0 = -\frac{a_3}{\sqrt{k_c}}\sqrt{2gy_3^0} + (1-\gamma_1) k_1 u_1^0$$
$$0 = -\frac{a_4}{\sqrt{k_c}}\sqrt{2gy_4^0} + (1-\gamma_2) k_2 u_2^0$$

Finally, we get:

$$0 = -a_1\sqrt{2gy_1^0} + a_3\sqrt{2gy_3^0} + \gamma_1 k_1 u_1^0\sqrt{k_c} \tag{2.5}$$
$$0 = -a_2\sqrt{2gy_2^0} + a_4\sqrt{2gy_4^0} + \gamma_2 k_2 u_2^0\sqrt{k_c} \tag{2.6}$$
$$0 = -a_3\sqrt{2gy_3^0} + (1-\gamma_1) k_1 u_1^0\sqrt{k_c} \tag{2.7}$$
$$0 = -a_4\sqrt{2gy_4^0} + (1-\gamma_2) k_2 u_2^0\sqrt{k_c} \tag{2.8}$$

2.1.3. *Linearize the system.* We want to obtain the equivalent matrices A_{eq}, B_{eq} at equilibrium. For this we use the formulas: $A = \frac{\partial f}{\partial h}|_{h=h^0}$ and $B_{eq} = \frac{\partial f}{\partial u}|_{u=u^0}$ where f describes the system of equation (1.1), (1.2), (1.3), (1.4). Equations for the linearization can be found below:

$$\frac{\partial f}{\partial h_1}|_{h_1=h_1^0} = \begin{bmatrix}-\frac{a_1}{A_1}\frac{\sqrt{g}}{\sqrt{2h_1^0}} & 0 & 0 & 0\end{bmatrix}^T$$
$$\frac{\partial f}{\partial h_2}|_{h_2=h_2^0} = \begin{bmatrix}0 & -\frac{a_2}{A_2}\frac{\sqrt{g}}{\sqrt{2h_2^0}} & 0 & 0\end{bmatrix}^T$$
$$\frac{\partial f}{\partial h_3}|_{h_3=h_3^0} = \begin{bmatrix}\frac{a_3}{A_1}\frac{\sqrt{g}}{\sqrt{2h_3^0}} & 0 & -\frac{a_3}{A_3}\frac{\sqrt{g}}{\sqrt{2h_3^0}} & 0\end{bmatrix}^T$$
$$\frac{\partial f}{\partial h_3}|_{h_3=h_3^0} = \begin{bmatrix}0 & \frac{a_4}{A_2}\frac{\sqrt{g}}{\sqrt{2h_4^0}} & 0 & -\frac{a_4}{A_4}\frac{\sqrt{g}}{\sqrt{2h_4^0}} & 0\end{bmatrix}^T$$

Using $T_i = \frac{a_i}{A_i}\sqrt{\frac{2h_i^0}{g}}$ and noticing that $\frac{a_i}{A_j}\sqrt{\frac{2h_i^0}{g}} = T_i\frac{A_j}{A_i}$, we put everything together and obtain A_{eq}

$$A_{eq} = \begin{bmatrix}\frac{-1}{T_1} & 0 & \frac{A_3}{A_1 T_3} & 0\\ 0 & \frac{-1}{T_2} & 0 & \frac{A_4}{A_2 T_4}\\ 0 & 0 & \frac{-1}{T_3} & 0\\ 0 & 0 & 0 & \frac{-1}{T_4}\end{bmatrix}$$

We do the same for B_{eq}

$$
\begin{aligned}
\frac{\partial f}{\partial u_1}|_{u_1=u_1^0} &= \begin{bmatrix} -\frac{\gamma_1 k_1}{A_1} & 0 & 0 & -\frac{(1-\gamma_1)k_1}{A_4} \end{bmatrix}^T \\
\frac{\partial f}{\partial u_2}|_{u_2=u_2^0} &= \begin{bmatrix} 0 & -\frac{\gamma_2 k_2}{A_2} & -\frac{(1-\gamma_2)k_2}{A_3} & 0 \end{bmatrix}^T
\end{aligned}
$$

Putting all this together, we get B_{eq}:

$$
B_{eq} = \begin{bmatrix} \frac{\gamma_1 k_1}{A_1} & 0 \\ 0 & \frac{\gamma_2 k_2}{A_2} \\ 0 & \frac{(1-\gamma_2)k_2}{A_3} \\ \frac{(1-\gamma_1)k_1}{A_4} & 0 \end{bmatrix}
$$

defining $x = \begin{bmatrix} h_1 & h_2 & h_3 & h_4 \end{bmatrix}$ we end up getting the asked system:

$$
\begin{aligned}
\frac{dx}{dt} &= A_{eq}x + B_{eq}u \\
y &= \begin{bmatrix} k_c & 0 & 0 & 0 \\ 0 & k_c & 0 & 0 \end{bmatrix} x = Cx
\end{aligned}
$$

2.1.4. *Transfer function.* The transfer function from the precedent system can be computed using the formula $G(s) = C(sI - A)^{-1}B$. We got $sI - A_{eq} = \begin{bmatrix} s + \frac{1}{T_1} & 0 & -\frac{A_3}{A_1 T_3} & 0 \\ 0 & s + \frac{1}{T_2} & 0 & -\frac{A_4}{A_2 T_4} \\ 0 & 0 & s + \frac{1}{T_3} & 0 \\ 0 & 0 & 0 & s + \frac{1}{T_4} \end{bmatrix}$ It is quite easy to see that the inverse of this matrix will be $(sI - A_{eq})^{-1} = \begin{bmatrix} \frac{1}{s+\frac{1}{T_1}} & 0 & \frac{\frac{A_3}{A_1 T_3}}{(s+\frac{1}{T_1})(s+\frac{1}{T_3})} & 0 \\ 0 & \frac{1}{s+\frac{1}{T_2}} & 0 & \frac{\frac{A_4}{A_2 T_4}}{(s+\frac{1}{T_2})(s+\frac{1}{T_4})} \\ 0 & 0 & \frac{1}{s+\frac{1}{T_3}} & 0 \\ 0 & 0 & 0 & \frac{1}{s+\frac{1}{T_4}} \end{bmatrix}$ and for more visibility: $(sI - A_{eq})^{-1} = \begin{bmatrix} \frac{T_1}{T_1 s+1} & 0 & \frac{\frac{A_3 T_1}{A_1}}{(T_1 s+1)(T_3 s+1)} & 0 \\ 0 & \frac{T_2}{T_2 s+1} & 0 & \frac{\frac{A_4 T_4}{A_2}}{(T_2 s+1)(T_4 s+1)} \\ 0 & 0 & \frac{T_3}{T_3 s+1} & 0 \\ 0 & 0 & 0 & \frac{T_4}{T_4 s+1} \end{bmatrix}$ We then multiply by C to obtain $C(sI - A_{eq})^{-1} = \begin{bmatrix} \frac{c_1 A_1}{1+sT_1} & 0 & \frac{c_1 A_3}{(T_1 s+1)(T_3 s+1)} & 0 \\ 0 & \frac{c_2 A_2}{1+sT_2} & 0 & \frac{c_2 A_4}{(T_2 s+1)(T_4 s+1)} \end{bmatrix}$ Finally multiplying per B_{eq}, we obtain:

$$
G(s) = \begin{bmatrix} \frac{\gamma_1 k_1 c_1}{1+sT_1} & \frac{c_1(1-\gamma_2)k_2}{(T_1 s+1)(T_3 s+1)} \\ \frac{c_2(1-\gamma_1)k_1}{(T_2 s+1)(T_4 s+1)} & \frac{\gamma_2 k_2 c_2}{1+sT_2} \end{bmatrix} \tag{2.9}
$$

2.1.5. *Minimum phase or not minimum phase? That is the question.* Analysing the zeros of the transfer function is equivalent to analysing them of det $G(s)$, which is equivalent to analyse those of $(1+T_3s)(1+T_4s) - \frac{(1-\gamma_1)(1-\gamma_2)}{\gamma_1\gamma_2}$ We got:

$$\begin{aligned}(1+T_3s)(1+T_4s) - \frac{(1-\gamma_1)(1-\gamma_2)}{\gamma_1\gamma_2} &= \frac{1}{\gamma_1\gamma_2}(\gamma_1\gamma_2(1+T_3s)(1+T_4s) - (1-\gamma_1)(1-\gamma_2)) \\ &= \frac{1}{\gamma_1\gamma_2}(s^2(\gamma_1\gamma_2T_3T_4) + s\gamma_1\gamma_2(T_3+T_4) + (\gamma_1+\gamma_2-1)\end{aligned}$$

From order 2 polynomial, we get that the two zeros z_1 and z_2 which are the roots of this polynomial follow these equations:

$$\begin{aligned}z_1 + z_2 &= -\gamma_1\gamma_2(T_3+T_4) < 0 \\ z_1z_2 &= \gamma_1 + \gamma_2 - 1\end{aligned}$$

We deduce then that if z_1 and z_2 have the same signs, they must be both negative following the first equation. The second one tells us that they have the same sign iff $\gamma_1+\gamma_2-1>0 \Leftrightarrow \gamma_1+\gamma_2>1$. We conclude that there is no RHP zeros if $2 \geq \gamma_1+\gamma_2 > 1$ and that the system is minimal phase. On the other hand, if we have $1 \geq \gamma_1+\gamma_2 > 0$, the system has one RHP zero and is nonminimal phase.

2.1.6. *RGA.* RGA is given by the following $RGA(s) = G(s) \odot G^{-T}(s)$ Hence, we get

$$\begin{aligned}RGA(s) &= \frac{1}{\det G(s)}\begin{bmatrix} g_{11}(s) & g_{12}(s) \\ g_{21}(s) & g_{22}(s)\end{bmatrix} \odot \begin{bmatrix} g_{22}(s) & -g_{21}(s) \\ -g_{12}(s) & g_{11}(s)\end{bmatrix} \\ &= \frac{1}{\det G(s)}\begin{bmatrix} g_{11}(s)g_{22}(s) & -g_{12}(s)g_{21}(s) \\ -g_{21}(s)g_{12}(s) & g_{22}(s)g_{11}(s)\end{bmatrix} \\ \Rightarrow RGA(0) &= \frac{1}{\gamma_1\gamma_2c_1c_2k_1k_2}\left(\frac{1}{1-\frac{(1-\gamma_1)(1-\gamma_2)}{\gamma_1\gamma_2}}\right)\begin{bmatrix} \gamma_1\gamma_2c_1c_2k_1k_2 & (1-\gamma_1)(1-\gamma_2)c_1c_2k_1k_2 \\ (1-\gamma_1)(1-\gamma_2)c_1c_2k_1k_2 & \gamma_1\gamma_2c_1c_2k_1k_2\end{bmatrix} \\ &= \begin{bmatrix} \frac{\gamma_1\gamma_2}{\gamma_1\gamma_2-(1-\gamma_1)(1-\gamma_2)} & \frac{(1-\gamma_1)(1-\gamma_2)}{\gamma_1\gamma_2-(1-\gamma_1)(1-\gamma_2)} \\ \frac{(1-\gamma_1)(1-\gamma_2)}{\gamma_1\gamma_2-(1-\gamma_1)(1-\gamma_2)} & \frac{\gamma_1\gamma_2}{\gamma_1\gamma_2-(1-\gamma_1)(1-\gamma_2)}\end{bmatrix} \\ &= \begin{bmatrix} \lambda & 1-\lambda \\ 1-\lambda & \lambda\end{bmatrix}\end{aligned}$$

Hence for the min phase case and non min phase case we get:

$$\begin{aligned}RGA_{min}(0) &= \begin{bmatrix} 1.5625 & -0.5625 \\ -0.5625 & 1.5625\end{bmatrix} \\ RGA_{non\ min}(0) &= RGA_{min}(0)^T\end{aligned}$$

2.1.7. *Determining k_1 and k_2.* In this experiment, in order to determine the constants k_1 and k_2 we need to block the outlet of lower tanks. This will yield the relations described in Eq. 2.10.

$$\begin{cases} A\frac{dh}{dt} = q_{in} - q_{out} \\ q_{in} = \gamma k u \\ q_{out} = 0 \end{cases} \tag{2.10}$$

This gives the final relation in Eq. 2.11

$$k = \frac{A}{\gamma u}\frac{dh}{dt} \tag{2.11}$$

We can choose $u = 7.5\,V$ (50% of the max voltage), and we know the following constants: $A = 15.52\,cm^2$, $\gamma = 0.625$ for the minimum phase and $\gamma = 0.375$ for the non-minimum phase system. In order to determine the final values of the proportionality constants we need to experimentally determine the rate at which the tanks are filled up using the measurements y_1, y_2. On the more practical side, it is a good idea to make sure to let all the air out of the pipes to ensure steady water flow, and repeat the experiment a couple of times and take the average value for reliability. Thus:

$$\begin{cases} k_1 = 5.52 \\ k_2 = 5.45 \end{cases} \tag{2.12}$$

2.1.8. *Determining Effective Outlet Hole Areas.* Having obtained the k_1 and k_2 constants, we can use them to determine the outlet hole areas. We can bring the water tanks to their equilibrium levels h_i^0 $(i = 1, ..., 4)$ with the equilibrium u^0 input voltage. This way $\frac{dV}{dt} = q_{in} - q_{out} = 0$ where $q_{out} = a\sqrt{2gh^0}$ and $q_{in} = \gamma ku$ for the lower tanks and $q_{in} = (1-\gamma)ku$ for the upper ones. This way the effective outlet hole areas can be described by and calculated to be:

$$\begin{cases} a_1 = \frac{\gamma k_1 u^0}{\sqrt{2gh^0}} = 0.2469\, cm^2 \\ a_2 = \frac{\gamma k_2 u^0}{\sqrt{2gh^0}} = 0.2680\, cm^2 \\ a_{3,min} = \frac{(1-\gamma) k_2 u^0}{\sqrt{2gh^0}} = 0.1567\, cm^2 \\ a_{4,min} = \frac{(1-\gamma) k_1 u^0}{\sqrt{2gh^0}} = 0.1122\, cm^2 \end{cases} \tag{2.13}$$

Following the same procedure for the non-minimum phase system we obtain:

$$\begin{cases} a_{3,nonmin} = 0.3732\, cm^2 \\ a_{4,nonmin} = 0.3460\, cm^2 \end{cases} \tag{2.14}$$

2.2. **Manual Control.**

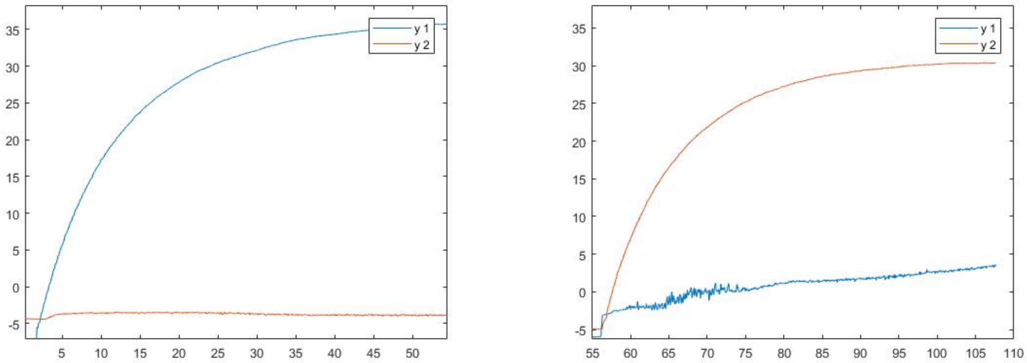
2.2.1. *Steady-state water levels (2.2.1).* The steady-state water levels are observed with a ruler when the pump is used at 7.5V. The expected values are computed from section 2.1.2 which are listed in the 1. The table contains water level measurements of both minimum and non-minimum phases. The differences are the upper tanks' holes and the input voltage for the non-minimum case is set to 69% for both pumps.

We can see that the measurements are different from the expected values which can have many practical causes like errors while measuring outlet areas and proportionality constants. The idea of steady-state might not be true because the level might be changing in tiny amounts. There are lots of irregularities that might have caused this. However, we can see two tanks in these processes are very close to the expected value.

2.2.2. *2.2.2.* The RGA that was derived before, for the minimum phase case, implies diagonal input-output coupling. That is, u_1 influences y_1 most directly and the same for u_2 and y_2. Since the off-diagonal values are non-zero then the inputs should be coupled. And indeed, in the minimum phase case we can see that as `Pump 1` is activated it mainly causes `Tank 1` water level to rise (Fig. 1(a)). The coupling is not as clear, most likely due to air in the pipes. The coupling is better visualized in Fig. 1(b) where `Pump 2` indirectly but visibly causes `Tank 1` water level y_1 to rise slightly together with the level y_2 in `Tank 2`. In the non-minimum phase case, the RGA matrix expects the off-diagonal input-output pairs to have the strongest influence on each other. However, the non-zero diagonal values still indicate coupling in the system. And indeed,

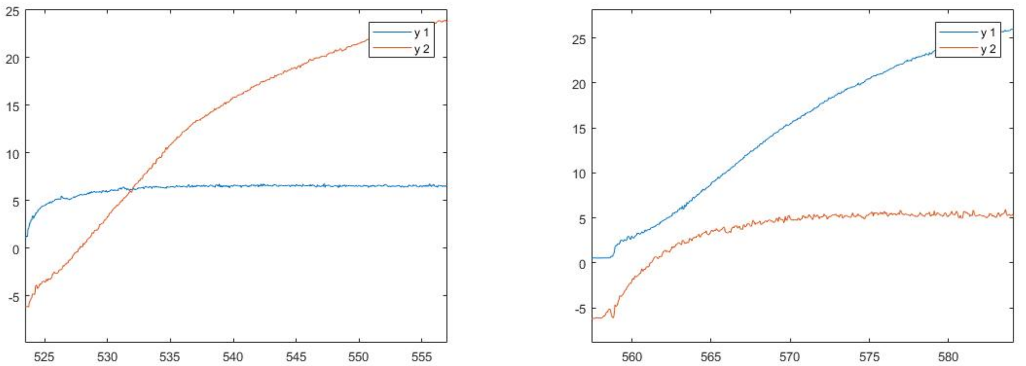
Tank	**Minimum Phase**		**Non-Minimum Phase**	
	Expected (cm)	**Measured (cm)**	**Expected (cm)**	**Measured (cm)**
Tank 1	14.33	7.3	27.29	10.9
Tank 2	11.85	10.6	22.58	6.9
Tank 3	5.02	11.9	1.68	7.9
Tank 4	9.54	15.6	1.90	5.4

TABLE 1. Comparison of expected and measured steady-state water heights in the four-tank process



((A)) Pump one. ((B)) Pump two.

FIGURE 1. Minimum phase.



((A)) Pump one. ((B)) Pump two.

FIGURE 2. Non-minimum phase.

u_1 mainly causes y_2 to change while also causing a small change in y_1 as shown in Fig. 2(a) (similarly for u_2 and y_1 in Fig. 2(b)). Overall conclusion: the system is coupled. Thus the theory agrees with the experiment findings.

2.2.3. *Transient times.* We tried to achieve 50% on both tanks by tweaking the input voltages for both minimum phase and non minimum phase. It took around 60s for minimum phase to achieve the reference while it took 80s for non-minimum phase to achieve the same. The mentioned values are the respective transient times.

2.2.4. *Important Differences.* The main difference is the setup of the plant for minimum and non-minimum case. The former is more intuitive to understand and easier to control.

3. Calculation of Controllers

Using the parameter values that were determined experimentally, the two controllers were calculated. The controllers used in the following section are a decentralized controller with dynamic decoupling and the robustified Glover-McFarlane version of that controller. The comparison of the two will allows us to see the differences in performance between the two in a real-life environment.

4. Laboratory Occasion 2

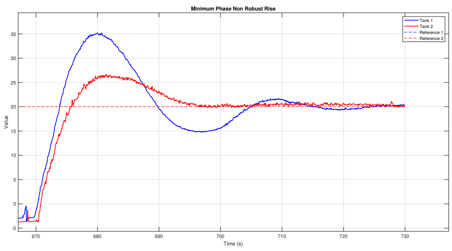


Figure 3. Minimum phase response to a step for decentralized controller

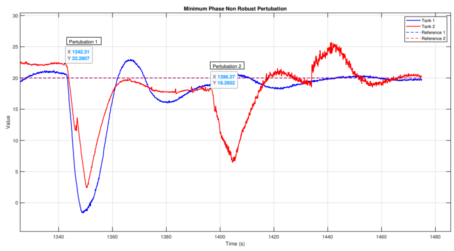


Figure 4. Minimum phase response perturbation for decentralized controller

4.1. **Decentralized Control.**

4.1.1. *Experiments on minimum and non-minimum phase cases.* As shown in Fig. 3 the decentralized dynamic decoupling controller handled the system reasonably well with respect to reference tracking. The rise time to references of value 20 were both approximately 10 seconds. The overshoot was greater for output 1- 75% compared to output 2- 25%. Furthermore, upon applying disturbance to the system (Fig. 4 we see that when both of the upper tank drains are opened and closed (`Perturbation 1` in Fig. 4), `Tank 2` drops in level. However, it comes back to the reference value after 60 seconds. Next, the system was exposed to `Perturbation 2`, also by opening one of the drains, resulting in similar a behavior. Lastly, the system was exposed to the last `Perturbation 3` (the spike at around $t = 1450s$ not labeled on the plot) in the form of

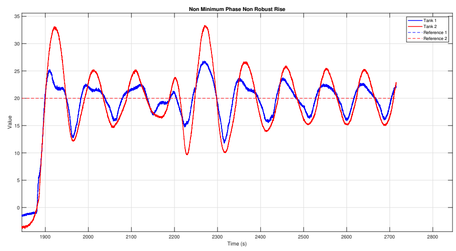


FIGURE 5. Non minimum phase rise time for decentralized controller

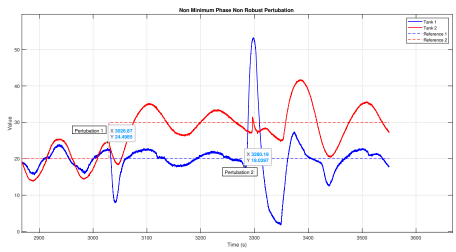


FIGURE 6. Non minimum phase perturbation for decentralized controller

pouring water into `Tank 1`. As expected, the water level rises sharply which is also combined with the up-tide of the sinusoidal oscillation. Then, the controller accounts for this disturbance and brings the level back to the reference. Admittedly, there should be more time in between perturbations to let the system settle back to steady state.

A similar experiment was conducted for the non-minimum phase system with a suitable controller. The controller did not manage to converge `Tank 1/2` to the reference values (`Reference 1` = `Reference 2` = 20) within a small margin (5% of the reference value) within the experiment time of 1700 s. It oscillates around 25% for `Tank 1` and 50% for `Tank 2` (Fig. 5). Nevertheless, the rise time is 10 s for both tanks. The overshoot is 25% for `Tank 1` and 65% for `Tank 2`. Next, the system is tested against disturbances as shown in Fig. 6. When exposed to `Perturbation 1` (opened drain), `Tank 1` level comes back to its oscillatory behavior while trying to approach the steady state. Thus the controller removes the extra disturbance brought upon the system. Similarly, when water is added (`Perturbation 2`), making `Tank 1` increase sharply, the controller manages to bring the level back to its "normal" behavior. Lastly, despite dynamic decoupling, we can still see coupling between the inputs and outputs as the disturbance in `Tank 1` also causes `Tank 2` level to increase.

4.1.2. *Differences in minimum and non-minimum phase case.* Comparing the two systems, with controllers relying on dynamic decoupling, we can see that the non-minimum phase system is more difficult to control. The oscillatory behavior would take a long time to approach the reference value which is a lot worse comparing to minimum phase case (Fig. 3. On the other hand, the rise time and overshoot values are similar in value.

4.2. **Robust Control.**

4.2.1. *Experiments on minimum and non-minimum phase cases.* Fig.7 shows the closed-loop response of the minimum-phase two–tank system under the robust controller designed via the Glover–McFarlane loop-shaping method. The step response exhibits a rise time of approximately

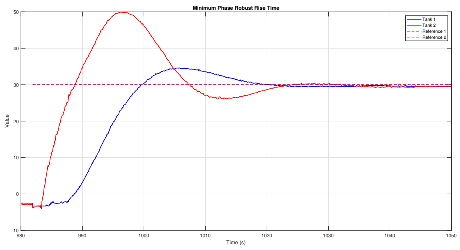


FIGURE 7. Minimum phase response to a step for robust controller

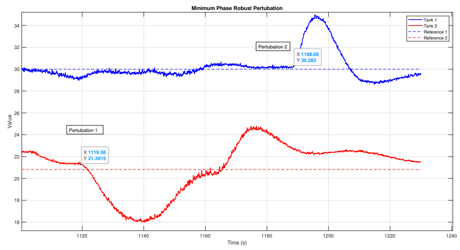


FIGURE 8. Minimum phase response perturbation for robust controller

4s and an overshoot of 20%, settling with a steady-state error of 2%. In the disturbance test (Fig.7), at $t = 1120$s a cup of water is poured into Tank2; the level returns to its reference in 40s (with no further overshoot beyond the original 20%) and holds within 2% of setpoint. Later, at $t = 1188$s another perturbation is introduced in tank 1 from 30 to 34.5: the system stabilizes in 20s, exhibits only a small oscillation of 3%, and settles within 1% steady-state error. As expected, inter-tank coupling is negligible throughout.

Fig.9 depicts the non-minimum-phase plant under the same Glover–McFarlane controller. The nominal step response has a rise time of 75s, an overshoot of 7%, and no measurable steady-state offset. A 10% reference change on Tank1 (30-33) at $t = 275$s settles in 150s without appreciable residual error.

Under disturbance (Fig.10), at $t = 460$s a cup of water is added to Tank1; the level returns to its setpoint in 15s but oscillates by 10% before damping out around 100s. Throughout both tests, coupling to Tank2 remains essentially zero, demonstrating the robust controller's ability to manage non-minimum-phase dynamics and reject disturbances with minimal interaction.

4.2.2. *Differences in performance of controllers.* The robust controller makes the system respond quicker to the steps introduced in the input, while the disturbance rejection is somewhat slower especially in the non-minimum phase case. Overall, the amplitude of oscillations due to the induced perturbations has significantly decreased.

4.2.3. *Differences in minimum and non-minimum phase case.* The non-minimum phase system was harder to control; this was evident from the above analysis. We can see that the response of a non-minimum phase system is slower than a minimum-phase system. However, disturbances are more influential in the non-minimum phase system.

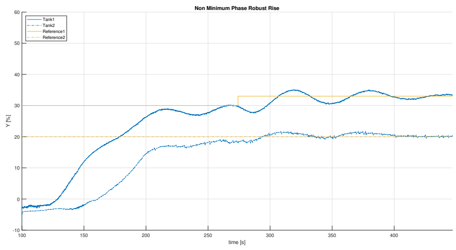


FIGURE 9. Non minimum phase rise time for robust controller

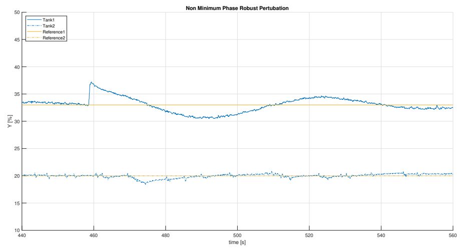


FIGURE 10. Non minimum phase perturbation for robust controller

5. CONCLUSION

In this project, we modelized a complex system with minimum and non minimum phase configuration. We computed the controllers to be able to command them both. We implemented a dynamic controller and a robust Glover-McFarlane controller on each. If there is not a lot of differences on the minimal phase case for both heuristics, the robust controller performs way better on the non minimal phase case, as it is more able to tackle the uncertainty of our modelisation.

REFERENCES

[1] Ljung, L. (1998). System identification. In Signal analysis and prediction (pp. 163-173). Boston, MA: Birkhäuser Boston.